Multicolor Multipartite Ramsey Theory:
A Combinatorial Design Approach

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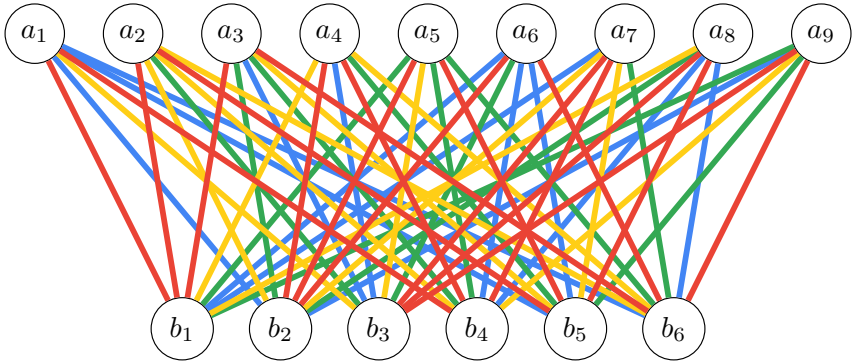
SoCal Discrete Math Symposium 2025

Multicolor Bipartite Ramsey Theory

Problem: What values for a and b guarantee that any edge coloring of $K_{a,b}$ with c colors contains an edge monochromatic subgraph isomorphic to $K_{s,t}$?

Multicolor Bipartite Ramsey Theory

A 4-coloring of the edges of $K_{9,6}$ has no edge monochromatic subgraph isomorphic to $K_{2,2}$.



Multicolor Multipartite Ramsey Theory

Problem: What values for $a_1, \ldots, a_d$ guarantee that any edge coloring of $K^{(d)}_{a_1,\ldots,a_d}$ with c colors contains an edge monochromatic subgraph isomorphic to $K^{(d)}_{s_1,\ldots,s_d}$?

Multicolor Multipartite Ramsey Theory

Simplification: Look for edge monochromatic hyperoctahedron $K^{(d)}_{2,\ldots,2}$.

Hyperoctahedron statistics can be used to establish quasirandom properties for sequences of hypergraphs due to Chung-Graham-Wilson Theorem (1989).

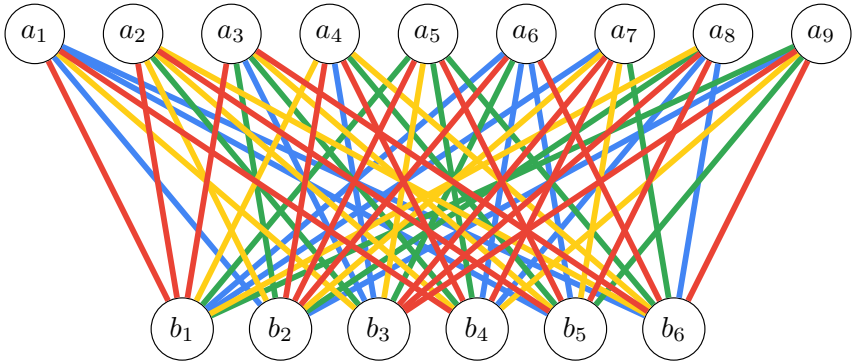
Multicolor Multipartite Ramsey Theory

Easy Lower Bound: Cooper-Fenner-Purewal (2008) apply Lovász Local Lemma

$$\prod_{i=1}^{d} a_i > \frac{c^{2^d-1}}{e2^d}$$

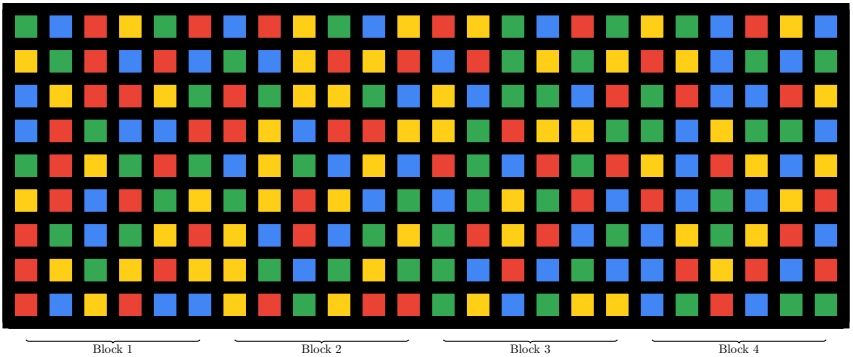
Digital Geometry

Get a new perspective by looking at the incidence matrix of the bipartite graph



Digital Geometry

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Digital Geometry

Reformulation for $K_{s,t}$: What are the dimensions of an $a \times b$ matrix whose entries are in $[c]$ such that there will always be a $s \times t$ sub-matrix with all the same entries?

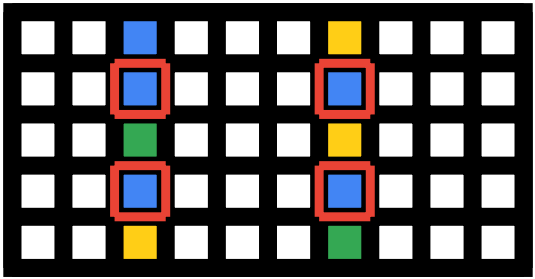
Reformulation for $K_{2,2}$: What values of a and b guarantee that any coloring $\chi : [a] \times [b] \to [c]$ contains a vertex monochromatic, grid-aligned rectangle?

Digital Geometry

Reformulation for $K^{(d)}_{2,\dots,2}$: What values of $a_1, \dots, a_d$ guarantee that any coloring $\chi : [a_1] \times \cdots \times [a_d] \to [c]$ contains a vertex monochromatic, grid-aligned hyper-rectangle (MC_d)?

Upper Bound Via Counting

A monochromatic rectangle (MC_2) appears when two columns have the same monochromatic pairs (MC_1).

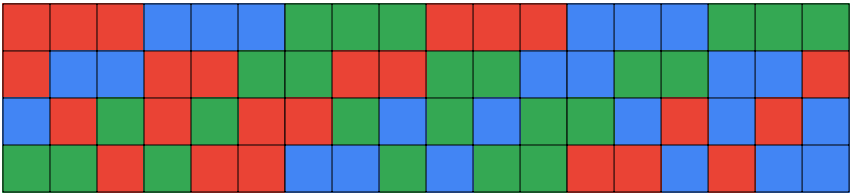


Upper Bound Via Counting

A grid with 4 rows and 3 colors must have at least one MC_1 per column by the pigeonhole principle. There are

$$3 \cdot \binom{4}{2} = 18$$

different ways an MC_1 can be picked from a column.



Upper Bound Via Counting

Let $\mathrm{mc}_d(a_1, \ldots, a_d; c)$ denote the minimum number of MC_d. If

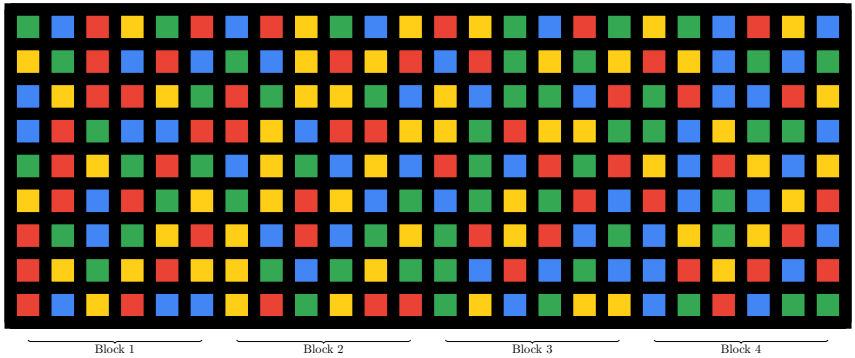
$$a_{d+1}\,\mathrm{mc}_d(a_1, \ldots, a_d; c) > c \prod_{i=1}^{d} \binom{x_i}{2}$$

then $\chi : [a_1] \times \cdots \times [a_{d+1}] \to [c]$ always contains an MC_d by the pigeonhole principle.

Remark: We can calculate $\mathrm{mc}_d(a_1, \ldots, a_d; c)$ for $d \geq 1$ when we know some specific constructions.

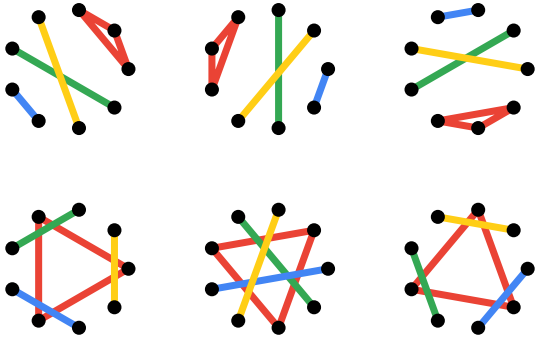
Lower Bound Via Constructions

Each column corresponds to a spanning union of sub-clique of a complete graph. A grid meeting the upper bound corresponds to a spanning clique decomposition of the complete graph.



Lower Bound Via Constructions

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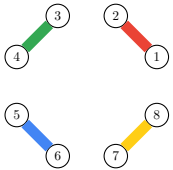


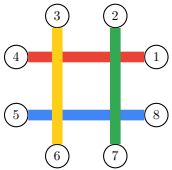
Decompositions and Designs

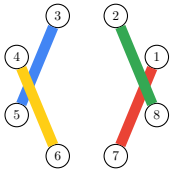
Spanning clique decompositions of the complete graph represent 2-$(v, K, 1)$ designs. We need resolvable designs in order to color the entire column.

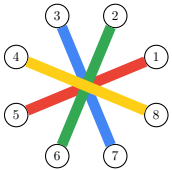
We can find constructions for the original multicolor bipartite problem (finding monochromatic $K_{s,t}$ instead of monochromatic $K_{2,2}$) using $s - (v, K, t)$ designs.

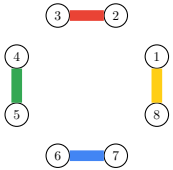
Resolvable K_2-Decompositions

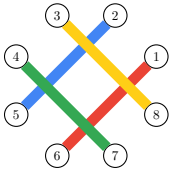


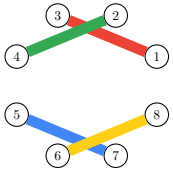




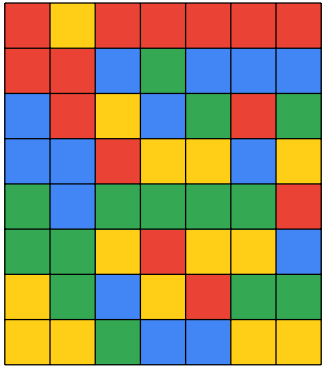




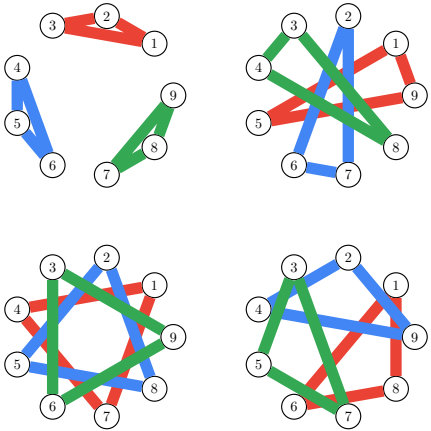




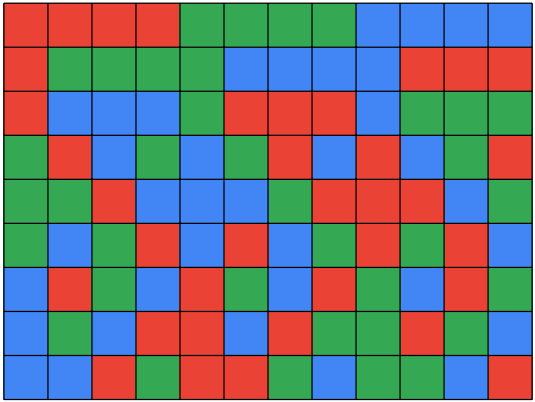
Resolvable K_2-Decompositions



Resolvable K_3-Decompositions



Resolvable K_3-Decompositions



Resolvable K_n Decompositions

Theorem. (Keevash 2018) Sufficiently large complete graphs admit resolvable clique decompositions whenever the obvious divisibility conditions are met.

Nonconstructive proof. Do not know what they look like for most cases.

Resolvable K_n Decompositions

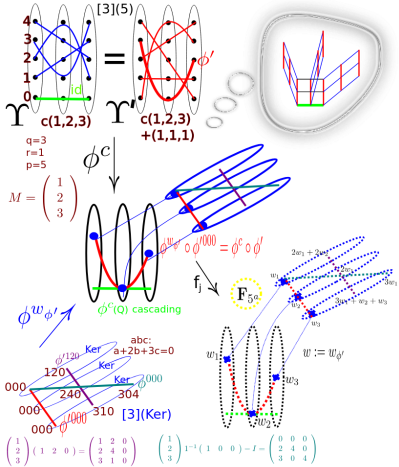


Figure 1: A cascade

Resolvable $\{K_n, K_{n+1}\}$ Decompositions

Theorem. (Lamken-Rees-Vanstone 1991) Resolvable $\rho K_2 + \tau K_3$ decompositions exist and there are some families that have been discovered.

Theorem. (Wei and Ge 2015) Resolvable $\rho K_3 + \tau K_4$ decompositions exist and there are some families that have been discovered.

Resolvable $\{K_n, K_{n+1}\}$ Decompositions

